

DANIEL A. AFERGAN

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PROFESSIONAL AND RESEARCH EXPERIENCE

Google Inc. Mountain View, CA & San Bruno, CA

Staff Software Engineer

May 2015 - Present

- YouTube Conversational AI & Personalized Recommendations *Jan. 2025 - Present*
 - Tech Lead / Manager, managing a cross-stack team to build and scale Generative AI components and LLM-powered recommendation products. Driving the rapid end-to-end execution of multiple novel AI products.
- YouTube User Understanding *Apr. 2021 - Jul. 2025*
 - Technical Lead for YouTube in-app surveys. Built and managed the end-to-end survey platform, including back-end infrastructure, service pipelines, and client-side UIs. Evolved the product from a simple A/B testing tool into a core signal provider for multiple ML recommendation models.
 - Senior Technical Lead for YouTube homepage recommendations ML infrastructure & pipelines. Oversaw reliability efforts, migrations, team goals, and engineering projects.
- YouTube Posts *Jan. 2017 - Apr. 2021*
 - Built infrastructure to introduce and scale lightweight content on YouTube Community Tab posts.
 - Developed new UIs and fan feedback mechanisms and researched their effects on engagement.
- Vanadium (v.io) & Fuchsia (fuchsia.dev) *May 2015 - Jan. 2017*
 - Created cross-device applications and researched automated UI design systems.
 - Built automated and crowd-powered Android app crawlers and UI datasets. Created the largest repository of mobile app designs (<https://www.interactionmining.org/archive/rico>).

Tufts University Human-Computer Interaction Lab Medford, MA

Research Assistant & Teaching Assistant

Jan. 2011 - May 2015

- Developed adaptive brain-computer interfaces (BCI) using functional near-infrared spectroscopy (fNIRS) as a passive input to improve performance.
- Built framework and optimized machine learning models to improve accuracy of real-time predictions of cognitive state
- Created Google Glass framework to adapt level of user notifications based on physiological state and accompanying 3D environment simulation testbed.

United States Naval Research Laboratory Washington, DC

Advisory Cognitive Scientist, Strategic Analysis Inc.

Jun. 2005 - Dec. 2010

- Provided scientific and technical analysis of virtual environments for automated training at the Warfighter Human System Integration Laboratory
- Researched, planned, and conducted experiments to test benefits of immersive locomotive virtual environment, augmented reality, and neurophysiological studies for US Marine Corps infantry training
- Created software to perform automated real-time mitigations of virtual environments according to physiological measurements, and correlate EEG signals with teamwork and military operational skills.

EDUCATION

Tufts University Medford, MA

PhD in Computer Science

May 2015

The George Washington University Washington, DC

Master of Science in Computer Science

May 2009

University of Pennsylvania Philadelphia, PA

Bachelor of Arts in Cognitive Science, concentration in Computation and Cognition

May 2005

Minors in Psychology and Computer Science and Engineering

SKILLS

Programming: Python, C++, Java, JavaScript, TypeScript, Go

- CONFERENCE PROCEEDINGS [1] Afergan, D. Using brain-computer interfaces for implicit input. *Proceedings of the adjunct publication of the 27th annual ACM symposium on User interface software and technology*, ACM Press, 2014.
- [2] Afergan, D., Hincks, S.W., Shibata, T. and Jacob, R.J.K. Phylter: a system for modulating notifications in wearables using physiological sensing. *International conference on augmented cognition*, Springer, 2015.
- [3] Afergan, D., Peck, E.M., Solovey, E.T., Jenkins, A.J., Hincks, S.W., Brown, E.T., Chang, R., and Jacob, R.J.K. Dynamic Difficulty Using Brain Metrics of Workload. *Proceedings of ACM Conference on Human Factors in Computing Systems (CHI) 2014*, ACM Press, 2014. **Best Paper Award Honorable Mention (top 5%)**.
- [4] Afergan, D., Shibata, T., Peck, E.M., Hincks, S.W., Yuksel, B.F., Chang, R., and Jacob, R.J.K. Brain-Based Target Expansion. *Proceedings of ACM Symposium on User Interface Software and Technology (UIST) 2014*, ACM Press, 2014.
- [5] Bailey, S.P., Pfluger, K.C., Holt, C., La Budde, Z., Afergan, D., Bartlett, S., Stripling, R., Miller, P.C., and Hall, E.E. Changes in Performance of a Virtual Reality Task Subsequent to Prolonged Exercise in the Heat and Carbohydrate Supplementation. *Proceedings of American College of Sports Medicine National Meeting*, 2006. Abstract published in *Medicine and Science in Sports and Exercise*, 38 (5 Supp.), S269-270.
- [6] Deka, B., Huang, Z., Franzen, C., Hirschman, J., Afergan, D., Li, Y., Nichols, J. and Kumar, R. Rico: A mobile app dataset for building data-driven design applications. *Proceedings of ACM Symposium on User Interface Software and Technology (UIST) 2017*, ACM Press, 2017.
- [7] Dong, T., Nebeling, M., Afergan, D., Churchill, E.F., Nichols, J., Goodman, E., Chi, P.Y.P., Li, Y. and Wigdor, D., 2016, June. The Making of Cross-Device Experiences: A Hands-on Workshop. *Proceedings of the 2016 ACM Conference Companion Publication on Designing Interactive Systems*, ACM Press, 2016.
- [8] Hincks S.W., Afergan D., and Jacob R.J.K. Using fNIRS for real-time cognitive workload assessment. *International Conference on Augmented Cognition 2016*, Springer, 2016.
- [9] Peck, E.M., Afergan, D., and Jacob, R.J.K. Investigation of fNIRS Brain Sensing as Input to Information Filtering Systems. *Proceedings of Augmented Human 2013*, 2013.
- [10] Sibert, L.E., Templeman, J.N., Stripling, R., Page, R.C., Coyne, J.T., La Budde, Z., and Afergan, D. Comparison of Three Virtual Environment Locomotion Interaction Techniques In Terms of Path Integration Performance. *Proceedings of Human Factors and Ergonomics Society Annual Meeting 2008*, 2008.
- [11] Stripling, R., Templeman, J.N., Sibert, L.E., Afergan, D., Cole, A., Cohn, J.V., Coyne, J.T., and La Budde, Z. Creating Effective First Person Training Tools: Evaluating Locomotion Interfaces. *Proc. American Psychological Association Conference 2005*, 2005.
- [12] Yuksel, B.F., Afergan, D., Peck, E.M., Griffin, G., Harrison, L., Chen, N.W., Chang, R. and Jacob, R.J.K.. Braahms: a novel adaptive musical interface based on users' cognitive state. *International Conference on New Interfaces for Musical Expression*, 2015.
- [13] Yuksel, B.F., Oleson, K.B., Harrison, L., Peck, E.M., Afergan, D., Chang, R. and Jacob, R.J.K. Learn piano with BACH: An adaptive learning interface that adjusts task difficulty based on brain state. *Proceedings of ACM Conference on Human Factors in Computing Systems (CHI) 2016*, ACM, 2016. **Best Paper Award (top 1%)**.

- BOOK
CHAPTERS
- [14] Afergan, D. and Davis, J.L. Promising Directions for Improved Training, Learning, and Memory. *Foundations of Augmented Cognition, 4th Ed.* Schmorow, D.D., Nicholson, D.M., Drexler, J.M., and Reeves, L.M. (Eds.) California: Falcon (2007), pp. 198-204. Presented at *Augmented Cognition International 2007*, 2007.
 - [15] Deka, B., Doosti, B., Huang, F., Franzen, C., Hibsichman, J., Afergan, D., Li, Y., Kumar, R., Dong, T. and Nichols, J., 2021. An Early Rico Retrospective: Three Years of Uses for a Mobile App Dataset. *Artificial Intelligence for Human Computer Interaction: A Modern Approach*, pp. 229-256. Springer, Cham.
 - [16] Peck, E.M., Afergan, D., Yuksel, B.F., Lalooses, F., Jacob, R.J.K. Using fNIRS to Measure Mental Workload in the Real World. *Advances in Physiological Computing*. Springer 2013.
 - [17] Stripling, R., Coyne, J.T., Cole, A., Afergan, D., Barnes, R.L., Rossi, K., Reeves, L., and Schmorow, D.D. Automated SAF Adaptation Tool (ASAT). *Foundations of Augmented Cognition, 3rd Ed.* Schmorow, D.D., and Reeves, L.M. (Eds.) Heidelberg, Germany: Springer-Verlag, pp. 346-353, 2011. Presented at *Proceedings of the Third International Conference on Foundations of Augmented Cognition*, 2007.
 - [18] Tognoli, E., Kovacs, A., Suutari, B., Afergan, D., Coyne, J.T., Gibson, G., Stripling, R., and Kelso, J.A.S. Behavioral and Brain Dynamics of Team Coordination Part I: Task Design. *Foundations of Augmented Cognition. Directing the Future of Adaptive Systems.* Schmorow, D.D. and Fidopiastis, C. (Eds.) Heidelberg, Germany: Springer-Verlag, pp. 257-264, 2011. Invited paper at *Human-Computer Interaction International 2011*.
 - [19] Tognoli, E., Kovacs, A., Suutari, B., Afergan, D., Coyne, J.T., Gibson, G., Stripling, R., and Kelso, J.A.S. Behavioral and Brain Dynamics of Team Coordination Part II: Neurobehavioral Performance. *Foundations of Augmented Cognition. Directing the Future of Adaptive Systems.* Schmorow, D.D. and Fidopiastis, C. (Eds.) Heidelberg, Germany: Springer-Verlag, pp. 376-382, 2011. Invited paper at *Human-Computer Interaction International 2011*.
- JOURNAL
ARTICLES
- [20] Bailey, S.P., Holt, C., Pfluger, K.C., La Budde, Z., Afergan, D., Stripling, R., Miller, P.C., and Hall, E.E. Impact of Prolonged Exercise in the Heat and Carbohydrate Supplementation on Performance of a Virtual Environment Task. *Military Medicine*, 173(2), pp. 187-192, 2008.
 - [21] Ottley, A., Peck, E.M., Harrison, L.T., Afergan, D., Ziemkiewicz, C., Taylor, H.A., Han, P.K. and Chang, R. Improving Bayesian reasoning: The effects of phrasing, visualization, and spatial ability. *IEEE transactions on visualization and computer graphics*, 22(1), pp. 529-538, 2016.
 - [22] Solovey, E.T., Afergan, D., Peck, E.M., Hincks, S.W., Jacob, R.J.K. Designing Implicit Interfaces for Physiological Computing: Guidelines and Lessons Learned using fNIRS. *ACM Transactions on Computer-Human Interaction (TOCHI)*, 2014.
- POSTER
PRESENTATIONS
- [23] Belyusar, D., Reimer, B., Mehler, B., Afergan, D., Coughlin, J.F., and Solovey, E.T. Utilizing functional near-infrared spectroscopy to identify cognitive processes contributing to workload in a dual-task environment. *Society for Neuroscience Annual Meeting*, 2014.
 - [24] Kovacs, A.J., Tognoli, E., Afergan, D., Coyne, J., Gibson, G., Stripling, R., and Kelso, J.A.S. Behavioral and brain dynamics of team coordination. *Society for Neuroscience Annual Meeting*, 2011.
 - [25] Kovacs, A.J., Tognoli, E., Afergan, D., Coyne, J., Gibson, G., Stripling, R., and Kelso, J.A.S. Brain dynamics of coordinated teams. *Society for Neuroscience Annual Meeting*, 2010.
 - [26] Shibata, T., Peck, E.M., Afergan, D., Hincks, S.W., Yuksel, B.F., and Jacob, R.J.K. Building Implicit Interfaces for Wearable Computers with Physiological Inputs: Zero Shutter Camera and Phylter. *Adjunct proceedings of ACM Symposium on User Interface Software and Technology (UIST) 2014*, ACM Press, 2014.

- [27] Yuksel, B.F., Peck, E.M., Afergan, D., Hincks, S.W., Shibata, T., Kainerstorfer, J., Tgavalekos, K., Sassaroli, A., Fantini, S., Jacob, R.J.K. Functional near-infrared spectroscopy for adaptive human computer interfaces. *SPIE Photonics West*, 2015.
- OTHER PAPERS
- [28] Afergan, D. Speed-Accuracy Comparison of Navigational Interfaces. Master's Thesis, The George Washington University, 2009.
- [29] Afergan, D., Peck, E.M., Chang, R., and Jacob, R.J.K. Using Passive Input to Adapt Visualization Systems to the Individual. *ACM CHI 2013 Workshop, Many People, Many Eyes: Aggregating Influences of Visual Perception on User Interface Design*, 2013.
- [30] Coyne, J.T., Stripling, R., Pfluger, K.C., LaBudde, Z., and Afergan, D. Company and Below Command and Control Information Exchange Study. *U.S. Naval Research Laboratory*, N0001406WX20812, 2007.
- [31] Crouser, R.J., Harrison, L., Afergan, D. and Peck, E.M., Beyond detection: investing in practical and theoretical applications of emotion+ visualization. *Proceedings of the 2016 EmoVis Conference on Emotion and Visualization*, pp. 35-38. Linkoping University, 2016.
- [32] Stripling, R., Templeman J.N., Sibert, L.E., Coyne, J.T., Page, R.G., La Budde, Z., and Afergan, D. Identifying Interface Limitations for Virtual Environment Training Systems. *Department of Defense Human Factors Engineering Technical Group Meeting 2006*, 2006.
- PATENTS
- [33] Drifting Keyboard. Tomoki Shibata, Robert J K Jacob, Daniel Afergan, Danielle Kong, U.S. Patent 15046584.
- [34] Automated Interface Design. Emily R Shack, Jeffrey W Nichols, Daniel A Afergan, Christian Robertson, U.S. Patent 15464279.